

Clinical Risk Assessment Report: Patient 25

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.41x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -2.3 [Bottom 2%] (-2.2%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 78.0 [Top 4%] (-1.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.59 [Top 5%] (-1.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.67 [Bottom 15%] (-0.9%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Carcinoembryonic Antigen (CEA) **Value: 2.1 [Avg (29th %ile)] (-0.7%)**
Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -1.24 [Bottom 3%] (-0.7%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.65 [Bottom 17%] (-0.7%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.65 [Avg (75th %ile)] (-0.6%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)

